No Spoiler Please! A TinyBERT Model to Detect Spoiler Reviews on IMDb

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Abstract

This report presents a knowledge distillation approach for spoiler detection in IMDb movie reviews. The goal is to transfer the predictive behavior of a fine-tuned BERT-base-uncased teacher model to a smaller and more efficient TinyBERT_General_4L_312D student model, while preserving competitive classification performance. To address the task, the teacher was first fine-tuned on the IMDb Spoiler Dataset using a head+tail truncation strategy for long reviews and a weighted cross-entropy loss to mitigate class imbalance. The student was then distilled in two phases: an intermediate-layer distillation phase based on embedding, hidden-state and attention alignment, followed by a prediction-layer distillation phase based on softened teacher logits and hard labels. Experimental results show that the distilled student remains close to the fine-tuned teacher, with only a 3.74% reduction in PR-AUC, while reducing the number of parameters by 86.76% and the inference time by 77.81%. Additional qualitative analyses on internal representations and attention patterns further suggest that the student successfully inherits part of the teacher's internal behavior. Overall, the results indicate that knowledge distillation is an effective strategy for obtaining a lightweight spoiler detector with a favorable efficiency–performance trade-off.

1 Introduction

Very often, before even seeing a film or a TV series, we decide to read some reviews because we're curious to know the opinions of those who have already seen it. In such a situation, it's not uncommon to encounter reviews that, whether intended to spoil something or not, include spoilers (e.g., "The movie is great, but character X dies/does this or that"). Clearly, we want to avoid this: luckily, platforms like IMDb warn us in advance about such reviews.

IMDb is by far the most popular global source for movies and TV shows, and it also includes user-submitted reviews with spoiler warnings for each one. The problem is that users mark reviews as spoilers themselves, but they can lie and insert spoilers on a no spoiler marked review. Other users can report a review as spoiler, and the IMDb moderation team will label it as such, but of course it's not an immediate action. What if there were a browser extension (running locally on users' PCs) that retrieves the text of a review and tells us "Spoiler alert!" before we can even begin reading it? With the recent evolution of LLMs, this task can be accomplished by fine tuning a sufficiently large pre-trained model, such as BERT.

Language model pre-training, such as BERT, has significantly improved the perfomances of many NLP tasks. However, pre-trained language models are usually computationally expensive, so it is difficult to efficiently execute them on resource-restricted devices (such as the user's PC). A novel Transformer distillation method, specially designed for knowledge distillation of the Transformer-based models, allows to effectively transfer to a smaller student model called TinyBERT the plenty of knowledge from a large pre-trained BERT model. [4]

1.1 Related Work

So far, three main studies have defined the paradigm of model compression via knowledge transfer.

- The first, by Hinton et al. [2], introduced the seminal concept of Knowledge Distillation (KD), focusing on transferring the dark knowledge from a large teacher to a compact student. Their approach minimizes the distance between the output probability distributions of the two networks, softened by a temperature T.
- The second study, by Jiao et al. [4], proposes a variant named TinyBERT trained into two distinct phases: General Knowledge Distillation first and Task-Specific Distillation second. While Hinton's method operates only on the final output layer, Jiao et al. argue that for complex language models, this is insufficient. Their method forces the student to mimic not only the teacher's prediction but also its intermediate representations.
- The third, by Ji & Zhu [3], attempts to provide a mathematical justification for why knowledge distillation works, demonstrating that the use of hard labels (ground truth) are necessary during distillation to counteract the so-called "imperfect teacher". This way, the student can understand how much he can trust the teacher's predictions.

The proposed TinyBERT model diversifies by using only task-specific distillation and introducing a loss factor on hard labels for prediction layer distillation.

2 Research question and methodology

Spoiler detection with LLMs is not a trivial task due to the subtle, context-dependent nature of what constitutes a spoiler, as well as the length of the reviews which can exceeding BERT's 512-token limit.

Research question: *To what extent can a highly compressed language model (TinyBERT) maintain not only the predictive performance of a larger teacher model (BERT-base) but also its internal reasoning and contextualization capabilities in a highly ambiguous task like spoiler detection?*

2.1 Proposed approach

The proposed approach is based on the following pipeline.

- **Data Preprocessing:** a truncation strategy that considers only the first 128 and last 382 tokens of a review's text was applied. This is because spoilers are often found at the end of a review, but the initial part is important because it introduces context. [6]
- **Teacher Fine Tuning:** layer-wise learning rate decay was used to avoid catastrophic forgetting. Also, weighted cross-entropy loss was used due to the dataset's imbalance.
- **Task-Specific Distillation:** this phase is divided into two subphases. In the first, intermediate-layer distillation is performed, while in the second, prediction-layer distillation is performed.
- **Evaluation and computational saving metrics:** some standard plots such as bar charts of evaluation metrics and confusion matrix have been constructed, as well as other plots to evaluate the computational savings of the student model compared to the teacher model.
- **Embeddings and Attention Matrices exploration:** in order to see if the student's internal representations actually aligned with those of the teacher, techniques such as PCA on the embeddings of both models and plots of the attention matrices using heatmaps were used.

3 Experimental Results

3.1 Data and Preprocessing

The dataset used was the IMDB Spoiler Dataset [5], available on Kaggle. This dataset (in JSON format) contains over 500,000 IMDb movie reviews, each with a boolean label that marks it as a spoiler or not. I choose to extract a random sample of 300,000 total reviews, then divided it into training (80%), validation and test sets (10% each). The dataset split was performed by grouping samples according to the *movie_id* attribute, ensuring that reviews of the same film do not appear in different splits and thus preventing data leakage.

To overcome the 512 maximum token limit with BERT, a solution based on the head+tail of the review text was used [6]: the text is first tokenized using the BERT vocabulary, also adding the special tokens [CLS] at the start and [SEP] at the end. If the resulting token list is longer than 512, only the first 128 tokens after [CLS] and the last 382 before [SEP] are retained; otherwise the sequence keeps its natural length and is padded dynamically at batch time. Each row, saved in a .parquet file for efficiency, contains *input_ids*, *attention_mask* and *token_type_ids*.

3.2 Teacher and Student Models Architectures

The choosen teacher model was BERT-base-uncased [1], while the choosen student model was TinyBERT_General_4L_312D [4]: this because this is exactly the TinyBERT model from Jiao et al with General Knowledge Distillation already performed and, more importantly, it allows to implement the same techniques they use to handle the mismatch in the number of layers and dimensions of hidden states and embeddings [4]. Table 1 shows the differences in the number of parameters between the two models.

Table 1: Parameters of the teacher and student models.

Model	Layers	Attention heads	Embeddings	Hidden states	Total params
BERT-base-uncased	12	12	768	768	110M
TinyBERT-general	4	12	312	312	14.5M

Both the fine tuning of the BERT teacher model and the task-specific distillation of the student were performed on a desktop PC equipped with an RTX 3070 (8GB of VRAM) and 32GB of RAM.

3.3 BERT Fine Tuning for Spoiler Detection

Since the dataset contains an imbalance in labels (73% no spoiler and 26% spoiler), a weighted cross entropy loss (WCE) was used during BERT fine tuning, giving more importance to the spoiler class. The weight has been calculated on the train set as the ratio between the number of negative samples and the number of positive samples.

The hyperparameters used for teacher fine tuning were the following:

- Epochs: 3
- Batch size: 24 (physical dimension = 8 and gradient accumulation = 3)
- Warmup ratio: 0.03
- Optimizer: AdamW with weight decay of 0.01 and a custom learning rate implemented as Layer-wise Learning Rate Decay (base LR = $2e-5$, decay factor = 0.95) to give a lower LR to the lower layers of the network. This, for BERT fine tuning, can overcome the catastrophic forgetting problem. [6]
- Metric for best model: PR-AUC on validation set. This choice is justified by the imbalance of the dataset and the chosen task.

Figure 1 shows the metrics obtained with the fine tuned teacher compared to the teacher without fine tuning.

3.4 Task-Specific Distillation

The task-specific distillation phase is in turn divided into two distinct sub-phases.

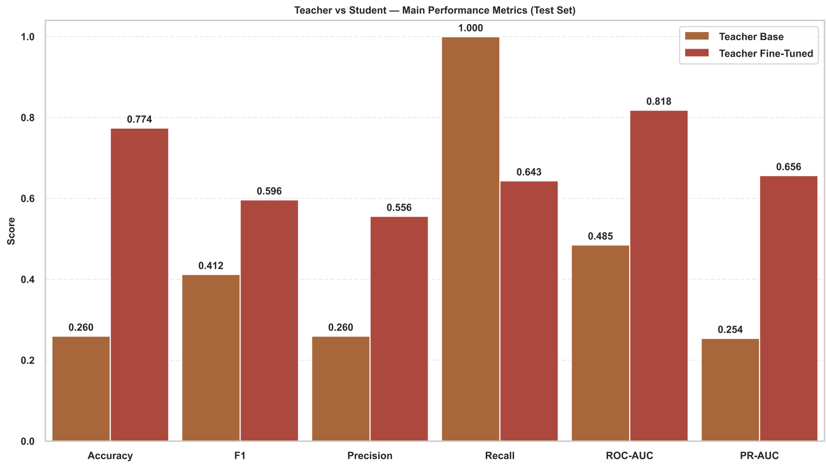

Figure 1: Evaluation metrics: BERT-base without and with fine tuning.

3.4.1 Phase 1: Intermediate Layers Distillation

During this subphase, embedding, hidden-state and attention distillation are performed, while the classification head is frozen on both models. This because attention weights learned by BERT can capture rich linguistic knowledge and is essential for natural language understanding, so it should be transferred to the student.

More formally, given an input x, the loss for this phase is calculated as

$$L_{TD1}(x) = \lambda_0 L_{embd}(x) + \sum_{m=1}^{M} \lambda_m (L_{hidn}^{(m)}(x) + L_{attn}^{(m)}(x)) \tag{1}$$

where

- $L_{attn} = \frac{1}{h}\sum_{i=1}^{h} MSE(A_i^S, A_i^T)$ is the attention-based distillation. h is the number of attention heads, $A_i \in \mathbb{R}^{l \times l}$ is the unnormalized attention matrix of the i-th head of teacher (A_i^T) or student (A_i^S) and l is the input text length (512 in this case).
- $L_{hidn} = MSE(H^S W_h, H^T)$ is the hidden states-based distillation. $H^S \in \mathbb{R}^{l \times d'}$ and $H^T \in \mathbb{R}^{l \times d}$ refer to the hidden states of the student (with size $d' = 312$) and teacher (with size $d = 768$) respectively, while $W_h \in \mathbb{R}^{312 \times 768}$ is a learnable linear transformation of the hidden states of student into the same space as the teacher's states. To ensure both teacher and student output pre-softmax attention logits, the forward pass of self-attention has been adapted accordingly.
- $L_{embd} = MSE(E^S W_e, E^T)$ is the embedding-based distillation. E^S and E^T refers to the student's and teacher's embeddings respectively, while W_e has the same role as W_h.

In this work, W_h and W_e were not treated as single matrices, but a separate linear projection $W^{(m)}$ for each matched representation has been learned.

To overcome the mismatch in the number of layers of student and teacher models, a mapping function $n = g(m)$ between student and teacher layers has been used. In this way, the $m-th$ layer of student model learns from the $g(m)-th$ layer of teacher model.

The hyperparameters used for this subphase were the following:

- Epochs: 8
- Batch size: 32 (physical dim = 8 and gradient accumulation = 4)
- Learning rate: $5e-5$

3.4.2 Phase 2: Prediction Layer Distillation

During this phase, only prediction layer distillation is performed in order to also fit the predictions of teacher model. A variant of prediction loss has been proposed here. Jiao et al. rely on the following loss defined by Hinton et al., which penalizes the soft cross-entropy loss between the student network's logits against the teacher's logits:

$$L_{pred} = CE(z^T/T, z^S/T) \tag{2}$$

However, by introducing the concept of imperfect teachers, it is possible to rewrite the loss by also introducing hard labels (ground truth) and weights for both soft and hard labels, so that the student learns from the teacher when the teacher produces correct predictions and learns to trust the teacher less when the teacher produces incorrect predictions.

The prediction layer loss implemented in this work has been the following:

$$L_{TD-2} = \rho(x) \cdot \underbrace{CE(z_T/T, z_S/T)}_{\text{Soft Loss}} \cdot T^2 + (1-\rho(x)) \cdot \underbrace{WCE(z_S, y_{true})}_{\text{Weighted Hard Loss}} \tag{3}$$

where

- $\rho(x)$: adaptive coefficient. For this work, $\rho(x) = 0.9$ has been used if teacher was correct, $\rho(x) = 0.2$ otherwhise.
- z_T, z_S : teacher's and student's logits respectively.
- T : the temperature parameter. For this work, $T = 1$ worked fine.
- $WCE(z_S, y_{true})$: Weighted Cross Entropy Loss between student's logits and ground truth. As in teacher's fine tuning, the weight has been calculated on the train set as the ratio between No spoiler and Spoiler reviews.

The hyperparameters used for this subphase were the following:

- Epochs: 3
- Batch size: 32 (physical dim = 16 and gradient accumulation = 2)
- Learning rate: $2e-5$
- Metric for best model: PR-AUC on validation set.

3.5 Results of distillation

As shown in this subsection, distillation achieved its goals.

3.5.1 Performance Drop

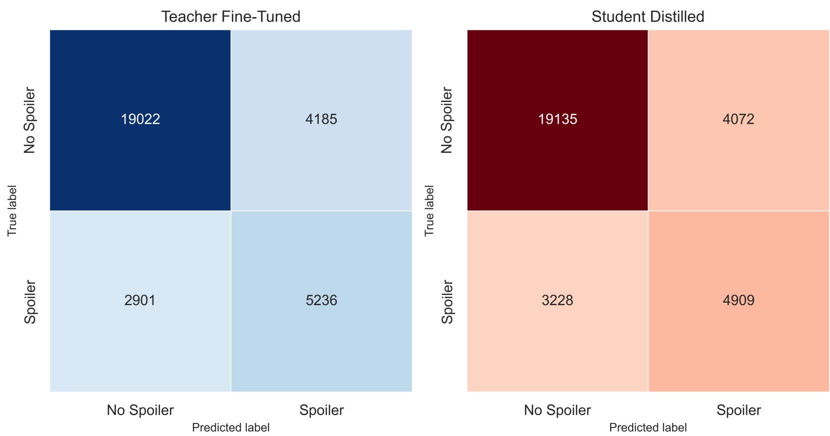

Figure 2: Confusion matrices: teacher fine tuned vs student distilled

Figure 2 and Figure **??** show the confusion matrices of the fine-tuned teacher and the distilled student, and the bar charts comparing the evaluation metrics of these two models with the COTS (out-of-the-shelf) student, respectively, all evaluated on the initially constructed test set. Among the reported metrics, ROC-AUC and PR-AUC are threshold-independent, whereas accuracy, precision, recall and F1 depend on the decision threshold selected on the validation set. It can be seen that the performance of the two models is very similar overall, with the distilled student showing a slight increase in false negatives and a corresponding small drop in recall. In particular, with respect to the fine-tuned teacher, the distilled student exhibits only a 3.74% decrease in PR-AUC, confirming that the performance loss introduced by KD is limited. The performance gap is similarly small across the other metrics, which supports the effectiveness of the proposed distillation strategy. The COTS student has a recall of 1, but all other metrics are much lower. This indicates that the model captures all actual positives, but at the cost of a high number of false positives. This behavior is expected in an imbalanced setting, where a small model may learn to predict the positive class too liberally.

3.5.2 Computational Savings

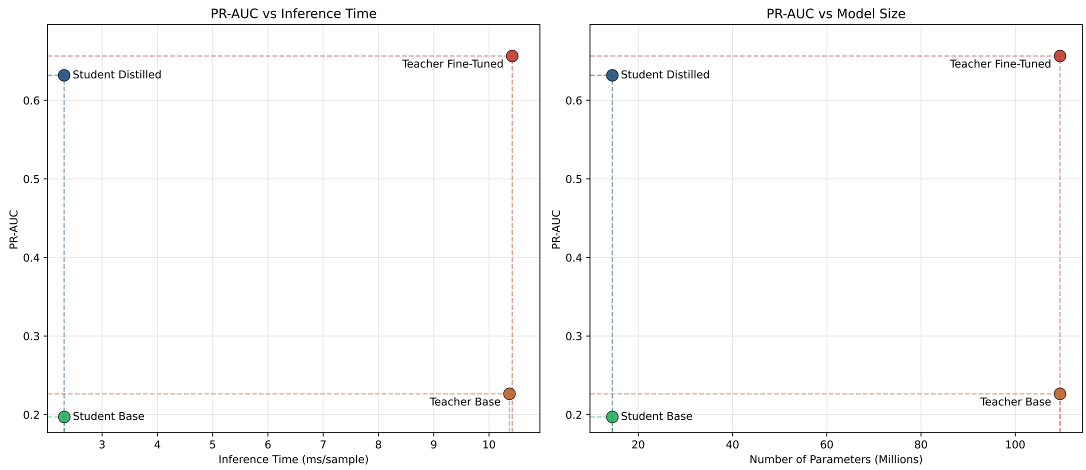

Figure 3: Scatter plots that compare PR-AUC and inference time (right) and PR-AUC and number of parameters (left) for all the models.

Figure 3 shows two scatter plots: the first one illustrates the relationship between PR-AUC and inference time (ms per sample), while the second one shows the relationship between PR-AUC and model size (# of parameters). Both graphs highlight that the distilled student lies in a substantially more favorable region of the efficiency–performance space than the teacher base model and the COTS student. In particular, compared to the fine-tuned teacher, it reduces inference time by 77.81% and the number of parameters by 86.76%, while remaining close in predictive performance. Therefore, the distilled student represents the best efficiency–performance compromise among the compared models.

3.5.3 Interpretability

Embeddings analysis

To qualitatively assess the alignment of internal representations after the intermediate-layer distillation phase, the last-layer hidden states of the fine-tuned teacher and of the phase-1 student were compared on a balanced subset of the test set. Since the student hidden size is smaller than the teacher hidden size, the student representations were first projected into the teacher space through the learned linear projection used during distillation, and then mean pooling was applied over the token dimension. A 2D PCA was fitted in the teacher space and both teacher and student representations were projected onto the same axes. Figure 4 shows the resulting plots. The two distributions exhibit a strong overlap for both the spoiler and no-spoiler classes, while the average cosine similarity between teacher and student representations remains very high in both cases (around 0.96). Although PCA only provides a qualitative low-dimensional projection,

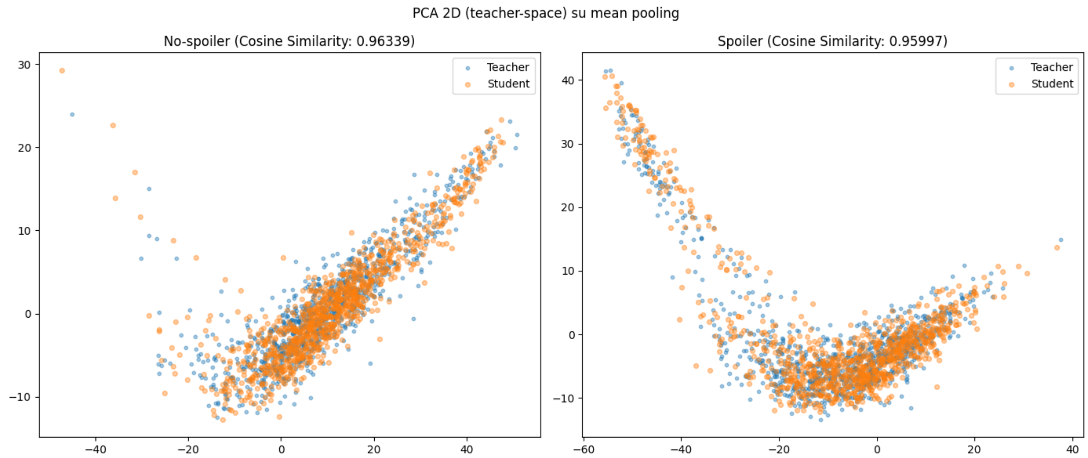

Figure 4: Inserisci qui la didascalia della tua figura.

these results suggest that the student learned to organize examples in a representation space that is highly consistent with that of the teacher.

Attention heatmap analysis

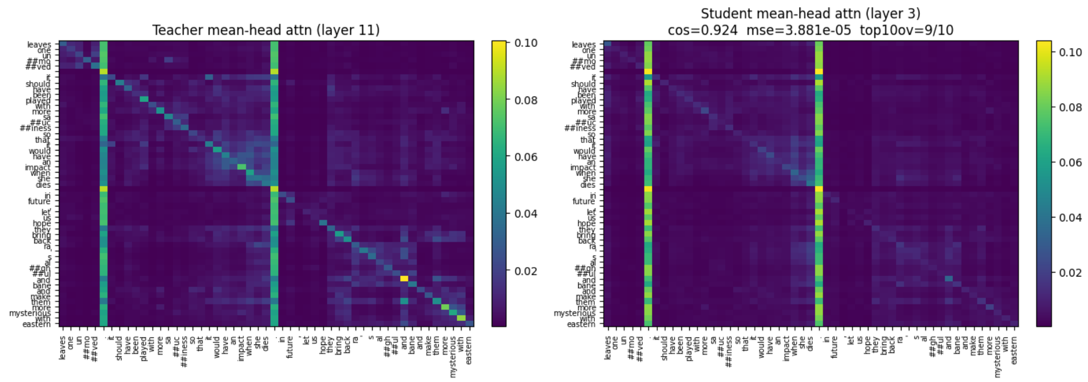

Figure 5: Inserisci qui la didascalia della tua figura.

To complement the embedding-level analysis, a qualitative inspection of attention patterns was performed on a manually selected challenging review containing the token *dies*, which is typically associated with spoiler-like content. For visualization purposes, the attention distributions of the last teacher layer and last student layer were averaged across heads and restricted to a local window centered around the target token. Figure 5 shows the resulting heatmaps. The two attention patterns exhibit a very similar structure, with high cosine similarity and very small mean squared error, and the overlap among the top-attended positions is also high. This indicates that the student focuses on contextual cues that are largely consistent with those used by the teacher, instead of reacting only to the presence of a potentially misleading keyword. The result should be interpreted as qualitative evidence on a single example, but it still supports the claim that intermediate-layer distillation helped the student inherit not only the teacher's predictions, but also part of its internal contextualization behavior.

4 Conclusions

References

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